

Chapter 27: Vaginal Delivery — Revision Notes

Williams Obstetrics 26e, pp. 1273–1323 of the PDF. High-yield numbers are in **bold**. The one table (27-1) is transcribed from the PDF image (it is missing from the extracted chapter text). 12 of the 13 figures are included (Fig 27-7, membrane teasing with ring forceps, is omitted).

Big picture: Spontaneous vaginal vertex delivery carries the lowest maternal risk. The goals are a controlled birth with minimal trauma, **prevention of OASIS**, fast and orderly management of **shoulder dystocia**, and a safe **third stage** (intact placenta, no inversion, no hemorrhage).

1. Delivery Technique

Preparation

- End of 2nd stage is heralded by **perineal bulging** and the scalp showing between the labia. Head pressure on the perineum triggers reflex pushing.
- Continue FHR monitoring: a **nuchal cord** often tightens with descent → deepening variable decelerations.
- Catheterize a distended bladder (more pelvic room).
- **Dorsal lithotomy** is most common. Stirrups vs no stirrups: **equal laceration rates** (RCT, n = 214).
 - Legs not spread too widely, not one higher than the other, and **never strapped in** → allows rapid thigh flexion (McRoberts) if shoulder dystocia develops.
- Vulvar/perineal cleansing, optional drapes; scrub, gown, glove, mask and eyewear.

Delivery of the head

- Position is usually **OA** or slightly oblique; **OP persists in ~5%**.
- **Crowning** = the vulvar ring encircles the largest head diameter.
- **OASIS** = 3rd- + 4th-degree lacerations (involve the anal sphincter ± perineal body/mucosa). **Preventing OASIS is a main goal.**

Perineal protection techniques

Technique	Effect
Antepartum perineal massage from 34–35 wk (1–2 fingers, 3 cm in; 2 min down, then 2 min to each side)	Nulliparas: ↓ episiotomy and ↓ 1st-degree tears. Does NOT ↓ 3rd/4th-degree tears
Epi-No intravaginal balloon (antepartum)	No benefit (episiotomy, perineal trauma, levator injury)
Intrapartum perineal massage (2nd stage)	One large RCT: no difference. Systematic reviews: ↓ OASIS , ↑ intact perineum
Hands-on (V of thumb/fingers bolsters perineum; other hand keeps head flexed and slows it) vs hands-poised (hands off unless expulsion is rapid)	Hands-poised → ↓ episiotomy . Intact perineum / OASIS rates not consistently different
Ritgen maneuver (chin pushed up through the perineum between contractions; other hand on occiput)	Expedites delivery, but does NOT change OASIS rates

- Once the introitus opens to **≥5 cm**, slow the head and control expulsive forces.

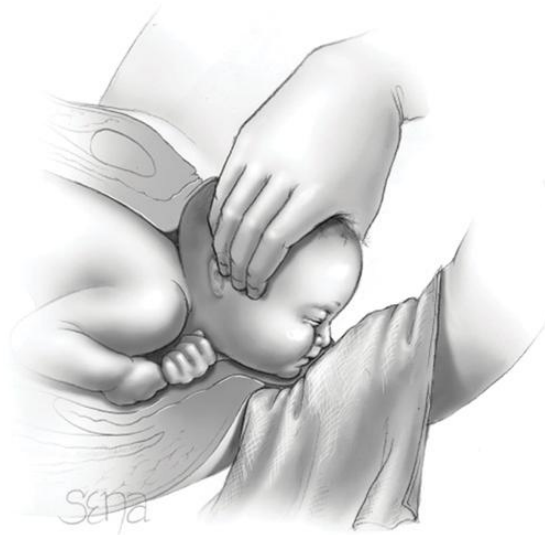


Figure 27-1. Modified Ritgen maneuver: upward pressure on the chin via the posterior hand while the occiput is held against the symphysis.

Delivery of the shoulders

- Feel for a **nuchal cord**: present in **~25% at term** (rises with GA).
 - Loose → slip over the head. Tight → cut between two clamps.
 - Tight nuchal cords occur in **~6%** of deliveries, with **no worse neonatal outcome**.
- **External rotation (restitution)**: occiput turns to a maternal thigh → the **bisacromial diameter** rotates into the AP diameter of the pelvis.
- If delayed: grasp the sides of the head and apply **gentle axial traction** (in line with the fetal spine); lateral neck deviation **≤25–45 degrees** → anterior shoulder under the pubic arch → then posterior shoulder.
 - Avoid abrupt/forceful lateral extension (**brachial plexus injury**).
 - **Don't hook fingers in the axillae** (upper-extremity nerve injury/paralysis).
- **Suctioning**: routine nasopharyngeal bulb suction is out (can cause **neonatal bradycardia**).
 - **Meconium**: no routine intubation/tracheal suction for **vigorous OR nonvigorous** neonates.
 - Suction only for **obvious airway obstruction** or when **PPV** is needed.

Cord clamping

- Two clamps **6–8 cm** from the abdomen; cord clamp later **2–3 cm** from the insertion.
- **Delayed clamping 30–60 s** for vigorous **term** neonates (baby held below the uterus).
 - Benefits: **↑ iron stores**, **↓ infant anemia** (especially where iron deficiency is common).
 - No worse Apgar or cord pH. **No ↑ maternal PPH** or change in postpartum Hb.
 - Harm: **↑ Hb** → **↑ hyperbilirubinemia** risk (phototherapy rates probably not different) → have jaundice protocols.
 - Cautions: need for urgent maternal/fetal resuscitation, abnormal placentation, abruption, FGR.
- **Preterm**: delayed clamping **≥30 s** → **↓ transfusion and ↓ death before discharge** (systematic reviews; one large RCT negative for its composite outcome).
- **ACOG**: **delayed clamping for vigorous term AND preterm** neonates not needing immediate resuscitation.
- **Cord milking**: safe at term (useful if rapid clamping is needed). In preterms it may **↑ severe IVH at <32 wk** → **not recommended routinely at <29 wk**.

Occiput transverse (OT)

- Persistent OT from pelvic shape: **platypelloid** (flat AP) or **android** (heart-shaped) pelvis.

- Without architectural abnormality or asynclitism, OT is **transient** and usually rotates to OA unless contractions are hypotonic.
- Aids: **manual rotation** (to OA, less often OP) or **Kielland forceps** (Ch 29).

2. Persistent Occiput Posterior Position

- **2–10%** of singleton, term, cephalic fetuses deliver OP. Many are OA early and malrotate in labor.
- **Risk factors:** **epidural**, nulliparity, larger fetal weight, prior OP delivery, **anthropoid pelvis**, **narrow subpubic angle**.
- **Maternal consequences:** prolonged 2nd stage, ↑ cesarean, ↑ operative vaginal delivery, ↑ blood loss and ↑ **OASIS**.
- **Neonatal:** ↑ acidemic cord gases, birth trauma, **Apgar <7**, NICU admission.
- **Diagnosis:** digital exam can be inaccurate → **transabdominal sonography** (transducer transverse above the mons): orbits and nasal bridge ventral, occiput against the lower sacrum.
- Maternal positioning (antepartum or in labor) does **not** ↓ persistent OP.
- **Management options:**
 1. Spontaneous OP delivery (roomy outlet, relaxed perineum in multiparas)
 2. **Manual rotation to OA** → ↓ cesarean, ↓ severe laceration, ↓ chorioamnionitis, ↓ blood loss
 3. Forceps rotation to OA (skilled operators)
 4. Forceps or vacuum delivery **from OP** (Ch 29)
- ⚠ **Scalp at the introitus can be misleading:** it may be **molding + caput**, not descent; the head may not even be engaged. Long labor, slow descent, head palpable above the symphysis → **prompt cesarean**.

3. Shoulder Dystocia

- The **anterior shoulder is wedged behind the symphysis** and fails to deliver with pushing and gentle axial traction.
- **Turtle sign** = head retracts against the perineum.
- An **emergency** (cord compressed in the birth canal) → team approach, clear leadership and communication.
- **Definition:** no consensus.
 - ACOG: **maneuvers required** to free the shoulder.
 - Spong: head-to-body time **24 s normal vs 79 s** with dystocia → proposed **>60 s** threshold.
 - In practice: clinical perception that normal traction fails.
- **Incidence ~1%** of deliveries; rising (↑ birthweight, better documentation).

Maternal and neonatal consequences

- Greater risk to the **fetus** than the mother.
- **Mother:** severe perineal tears, **PPH** (mainly **atony**, also lacerations).
- **Neonate:**

Outcome	Rate
Brachial plexus injury	11% (n = 1177)
Clavicular or humeral fracture	2%
Acidosis at delivery	7% (n = 514)
Cardiac resuscitation or HIE	1.5%
Severe acidosis / HIE if head-to-body <5 min	0.5% / 0.5%
Severe acidosis / HIE if ≥5 min	6% / 24%

Prediction and prevention

- **Cannot be accurately predicted or prevented** (ACOG).
- **Associations:** fetal **macrosomia**, maternal obesity, prolonged 2nd stage, operative vaginal delivery, **prior shoulder dystocia**. Risk rises with birthweight.
 - Macrosomia linked to maternal obesity, postterm, multiparity, diabetes.
 - **Macrosomia + diabetes** escalates risk: diabetic fetuses have larger shoulder/extremity circumferences and larger shoulder-to-head and chest-to-head ratios. Sonographic thresholds based on these have **poor sensitivity**.
- **Planned cesarean should be considered** (ACOG) if EFW:
 - **>5000 g without diabetes**
 - **>4500 g with diabetes**
- **Early induction (37–39 wk) for suspected macrosomia:**
 - Boulvain RCT (~800, insulin-requiring diabetics excluded): dystocia ↓ **by two thirds**, no brachial plexus injury in either group; ↑ hyperbilirubinemia/phototherapy with induction.
 - Meta-analysis of 4 RCTs: no difference in cesarean, dystocia or operative delivery; ↓ **fracture** with induction.
 - EFW is inaccurate; weigh against early-delivery harms. **ACOG: no induction <39 wk for suspected macrosomia alone.**
- **Recurrence: 4–13%.** Labor is still reasonable for many.
 - Evaluate EFW, GA, maternal glucose intolerance and severity of prior neonatal injury; discuss cesarean risks/benefits. **Either route is appropriate** after counseling.

Management

- **Two competing goals:** shorten delivery time (ongoing cord compression) vs avoid injury from aggressive manipulation.
- Call for help: nurses, obstetric providers, **anesthesia**. Adequate analgesia helps.
- **Episiotomy** may give room for maneuvers but **does NOT ↓ brachial plexus injury** itself.

Maneuver	How	Mechanism / note
McRoberts (usually first)	Legs out of stirrups, hips sharply flexed onto abdomen , knees flexed	Sacrum straightens vs lumbar spine, symphysis rotates cephalad , ↓ pelvic inclination. Does NOT enlarge the pelvis ; ↓ force needed to free the shoulder
Suprapubic pressure (with McRoberts)	Assistant pushes down and lateral	Abducts anterior shoulder (↓ bisacromial diameter) and rotates shoulders into the oblique diameter (longer than AP at the inlet)
Delivery of posterior arm	Hand along the posterior humerus; sweep the arm across the chest, elbow flexed; grasp hand and deliver along the face	Fingers parallel to the humerus ↓ fracture. ↓ bisacromial diameter, then rotate girdle into oblique
Rubin	Push the most accessible shoulder (often posterior) toward the fetal anterior chest	Abducts shoulders (↓ bisacromial diameter), rotates into oblique
Woods corkscrew	Press the anterior surface of the posterior shoulder toward the fetal back; rotate 180 degrees	Posterior shoulder becomes anterior and delivers beneath, not behind , the symphysis
Gaskin (all-fours)	Mother on hands and knees; axial traction frees posterior shoulder	Hard with regional analgesia; repositioning takes time
Posterior axilla sling traction	Suction tubing/stiff catheter looped under posterior axilla; pull up and out	For an inaccessible posterior arm; 96% success as first maneuver (n = 119)
Deliberate clavicle fracture	Thumb presses anterior clavicle against the pubic ramus	Hard in a large baby; heals fast; trivial vs plexus injury/asphyxia/death
Symphysiotomy	Divide symphyseal cartilage/ligaments under local; widens up to 2.5 cm ; displace urethral catheter laterally with a vaginal finger	Rarely used (training, maternal pelvic/urinary injury)
Zavanelli	Terbutaline 0.25 mg SC ; rotate head to OA/OP, flex, push back up → cesarean	Success 91% cephalic, 100% breech head entrapment; fetal injury common (prior manipulations)
Cleidotomy	Cut the clavicle with scissors	Usually for a dead fetus

- McRoberts, suprapubic pressure, posterior arm and rotational maneuvers have **similar neonatal injury rates**. **Longer duration and more maneuvers → more injury**. Maneuvers may be repeated.
- **Simulation drills** improve performance and documentation; neonatal outcome benefit inconsistent.

Memory hook: Order of escalation: "**McRoberts + Suprapubic → Posterior arm → Rotate (Rubin/Woods) → All-fours / sling → Last resorts (clavicle, symphysiotomy, Zavanelli, cleidotomy)**".



Figure 27-2. McRoberts maneuver with suprapubic pressure directed down and lateral.

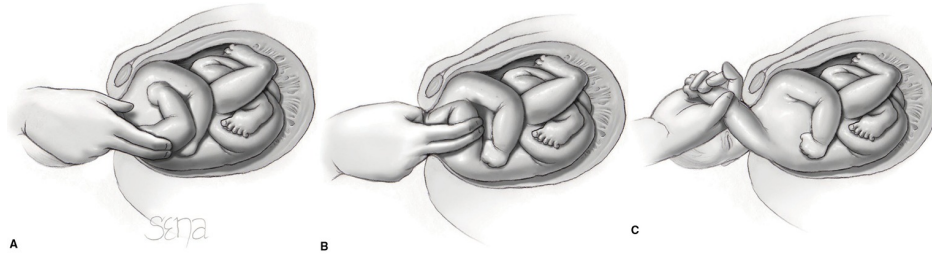


Figure 27-3. Delivery of the posterior arm: follow the humerus, sweep the flexed arm across the chest, then deliver it along the face.

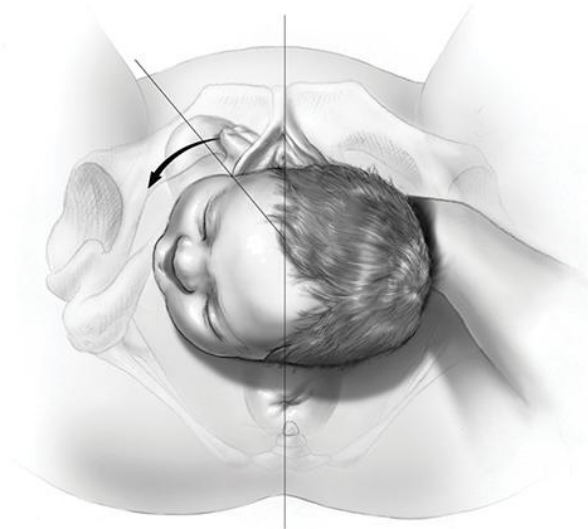


Figure 27-4. Rubin maneuver: push the accessible shoulder toward the fetal chest to abduct both shoulders and shrink the bisacromial diameter.

4. Special Populations

Home birth

- US 2017: **1.4% planned**, 0.2% unplanned.
- **Unplanned** (Norway): **1.1%** fetal/neonatal death (infection, prematurity, abruption). Risks: multiparity, distance from hospital (Norway); youth, no prenatal care, minority race, lower education (US).
- **Planned low-risk**: fewer interventions (augmentation, episiotomy, operative delivery, cesarean).
- **US midwife-attended home birth: neonatal mortality 1.3/1000, ~4×** that of midwife-attended hospital births. ↑ neonatal seizures and serious neurologic dysfunction.
- Substantial risk with preeclampsia, multifetal gestation, prior cesarean, breech.
- **ACOG absolute contraindications: multifetal gestation, prior cesarean, breech.** Hospitals/accredited birth centers are safest, but patient autonomy is respected.

Water birth

- **1st-stage immersion**: ↓ regional analgesia use, no added harm (Cochrane).
- **Birth in water**: cord **avulsion ~3/1000** (baby lifted out abruptly); fresh-water drowning, pneumonia/sepsis → strict sanitizing.
- **ACOG: "birth occur on land, not in water."**

Female genital mutilation (FGM)

- Federal crime in the US for girls **<18 y**; condemned by major organizations. Practiced in Africa, Middle East, Asia.
- Long-term: urogenital infection, chronic vulvar pain, dyspareunia, psychiatric sequelae (offer referral).

Table 27-1. World Health Organization Classification of Female Genital Mutilation

Type	Description
Type I	Partial or total removal of the clitoris and/or prepuce
Type II	Partial or total removal of the clitoris and the labia minora, with or without labia majora excision
Type III	Partial or total labial minora and/or majora excision, followed by fusion of the wound, termed infibulation , to cover and narrow the vagina. With or without clitoridectomy
Type IV	Pricking, piercing, incising, scraping, cauterization, or other injury to female genitalia

- Assess introital elasticity and caliber with a finger or swab.
- **Types I, II and IV → routine care. Type III** or extensive scarring → higher risk; discuss release **early antenatally**.
- **Defibulation** (= deinfibulation = anterior episiotomy): midline division of scar to reopen the vulva.
- FGM ↑ **perinatal morbidity by 1–2%** (WHO); ↑ prolonged labor, lacerations/episiotomy, PPH.
- Type III + defibulation → ↓ **cesarean** and ↓ **OASIS**. **WHO recommends defibulation**, antepartum or intrapartum (intrapartum works well).
- Technique: lidocaine (if no regional) → two fingers or clamp tips behind the fused shelf, in front of urethra and head → **midline incision** → raw edges sutured with rapidly absorbable suture.
- **Counseling**: cultural sensitivity (avoid "mutilated/assaulted" language). **Partial** defibulation exposes vagina and urethra; **total** extends to the clitoris. Urine stream and menses will change; new anatomy visible. Wound complications rare; dyspareunia uncommon; **sexual function scores usually improve**.

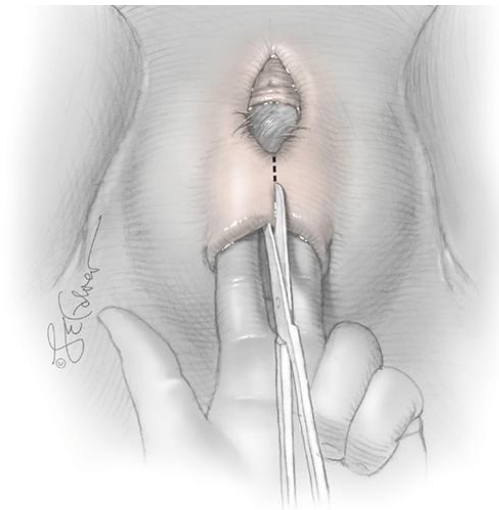


Figure 27-5. Deinfibulation: fingers protect the urethra and head while the fused shelf is incised in the midline.

Prior pelvic reconstructive surgery

Prior surgery	Delivery advice
Midurethral sling	Continence preserved in most (multiple pregnancies may ↑ recurrence). Vaginal delivery suitable
Sacral neuromodulation	Turn the device off in pregnancy. Lead displacement/efficacy loss after either route; vaginal birth not proscribed
Hysteropexy	Data too few; most have elective cesarean

Anomalous fetuses

- Obstruction can occur with extreme hydrocephaly, body-stalk anomaly, conjoined twins, massive abdominal enlargement (bladder, ascites, organomegaly).
- Mild hydrocephaly: vaginal delivery allowed if **BPD <10 cm** or **HC <36 cm**.

- If neonatal death has occurred or is certain: reduce size by sonographically guided **cephalocentesis/paracentesis**, or **cleidotomy**. Breech hydrocephalus → suprapubic cephalocentesis once the aftercoming head enters the pelvis.

5. Third Stage of Labor

Delivery of the placenta

- From birth of the fetus to delivery of the placenta. Goals: **intact placenta, no inversion, no PPH**.
- If the fundus is firm and bleeding minimal → **watchful waiting**. **No massage or fundal pressure** before separation; palpate often for atony.
- ⚠ **Never pull the cord to extract the placenta** → **uterine inversion**.
- **Signs of separation:**
 1. Sudden **gush of blood**
 2. **Globular, firmer fundus**
 3. **Cord lengthens** (moves outward)
 4. **Uterus rises** into the abdomen (placental bulk in the vagina pushes it up)
- **Median time to separation 4–12 min.**
- Expression: mother bears down; or, with a firmly contracted uterus, cord held slightly taut (not pulled), a hand around the fundus propels the placenta while the **heel of the hand pushes the uterus cephalad/toward the sacrum** (prevents inversion). Lift the placenta out; clamp and tease torn membranes.

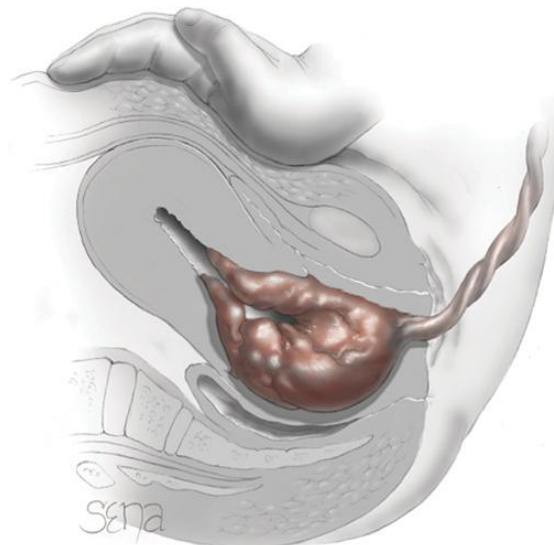


Figure 27-6. Expression of the detached placenta: the heel of the hand pushes the uterus cephalad while the cord is held in place, never pushing the fundus into the canal.

Expectant vs active management

	Components	Evidence
Expectant	Wait for separation; spontaneous delivery ± nipple stimulation or gravity	—
Active	(1) Early cord clamping, (2) controlled cord traction , (3) immediate prophylactic uterotonic	Bundle is questioned: delayed clamping doesn't ↑ PPH; cord traction doesn't prevent PPH but ↓ manual removal and shortens 3rd stage . Uterotonic is the essential component

- Post-placental **uterine massage** supported, though evidence against PPH is weak.
- Uterotonic **before or after** placental expulsion: timing doesn't affect PPH, retention or 3rd-stage length.
- **WHO: oxytocin is first line.**

Uterotonic agents

Agent	Dose / route	Key facts
Oxytocin (Pitocin)	20 U (2 mL) per 1 L crystalloid after the placenta at 10–20 mL/min (200–400 mU/min) until firm, then 1–2 mL/min until transfer. No IV: 10 U IM	Onset ~1 min , half-life 3–5 min ; store ≤25°C . IV bolus → profound hypotension (dangerous with hypovolemia/cardiac disease). Water intoxication with large volumes of electrolyte-free dextrose (antidiuretic effect). No standard dose established
Methylergonovine (Methergine)	200 µg IM , or slow IV over ≥60 s	↓ PPH and extra uterotonics; ↑ BP → contraindicated in hypertension . Only ergot made in the US (ergonovine similar)
Misoprostol (Cytotec, PGE ₁)	400 or 600 µg orally , single dose	Inferior to oxytocin ; for settings without oxytocin . S/E: fever, diarrhea
Carbetocin (Duratocin)	Long-acting oxytocin analogue	Equivalent to oxytocin after vaginal birth; more expensive; heat-stable form for low-resource settings
Oxytocin + ergonovine (Syntometrine)	—	Used outside the US
Tranexamic acid	1 g IV (studied)	Meta-analysis PPH 11.4 → 8.7%, but large RCT (n = 3891) no benefit → not recommended prophylactically (used for treatment, Ch 42/44)

Manual removal of placenta

- Placenta not promptly delivered in **~2%** of singletons. Causes:
 - Placenta adherens** (contractions insufficient to detach)
 - Lower uterine segment constriction** trapping a detached placenta
 - Placenta accreta spectrum**
- Risk factors**: stillbirth, prior cesarean, prior retention, **preterm delivery**.
- Time by which **90%** deliver spontaneously: **180 min at 20 wk; 21 min at 30 wk; 14 min at 40 wk**.
- PPH risk rises with 3rd-stage length. Without bleeding, evidence supports intervening at **15–20 min; WHO cites 60 min**.
- Brisk bleeding + undeliverable placenta** → **manual removal** (fundal hand + intrauterine hand sweeping side to side to cleave the placenta).
- Antibiotics**: evidence neither supports nor refutes (ACOG); **WHO recommends** prophylaxis. Authors give **single-dose cefazolin** (if not allergic/not already on antibiotics).

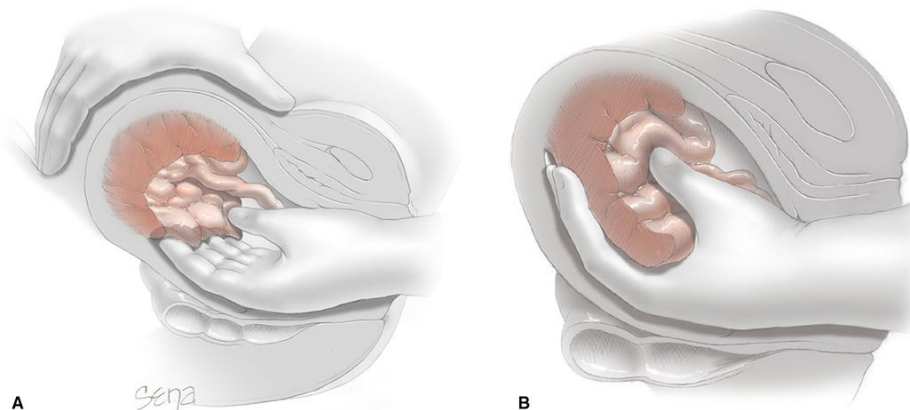


Figure 27-8. Manual removal: one hand on the fundus, the other sweeps the placenta off the wall and removes it.

6. Immediate Postpartum Care

- **The first hour is critical:** laceration repair, **atony most likely now**, hematomas may expand → check uterine tone and perineum often.
- **BP and pulse** immediately and **every 15 min for 2 h. Temperature every 4 h for 8 h**, then every 8 h. More often if high risk.
- Examine placenta, membranes and cord (Ch 6).
- **Skin-to-skin contact** (promotes breastfeeding):
 - Mother semi-reclined; dry the baby; chest on breast, legs flexed; warm blankets and cap.
 - Head uncovered, turned to the side, **sniffing position**, mouth and nares clear.
 - Monitor continuously for stability, somnolence, breathing and positioning (**smothering, falls**).

7. Lower Genital Tract Lacerations

Perineal laceration classification

Degree	Structures involved
1st	Vaginal epithelium or perineal skin only
2nd	Perineal muscles (bulbospongiosus, superficial transverse perineal); anal sphincter spared
3a	<50% of EAS torn
3b	>50% of EAS torn, IAS intact
3c	EAS + IAS torn
4th	Perineal body, entire sphincter complex and anorectal mucosa

- Cervical and vaginal lacerations → Ch 42.

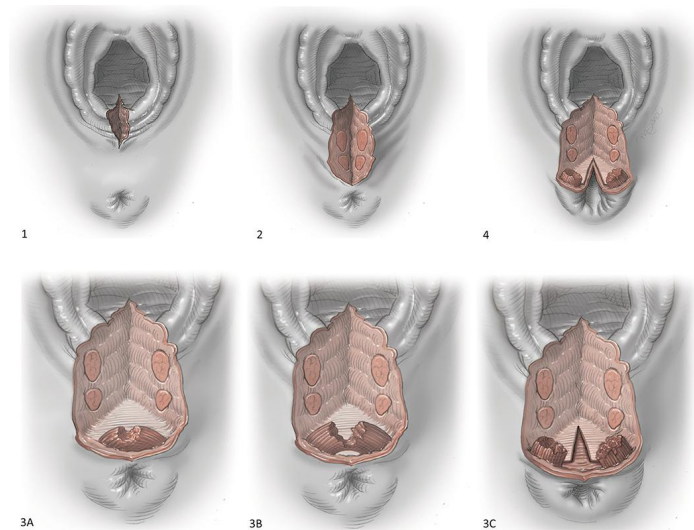


Figure 27-9. Perineal laceration degrees 1, 2, 3a, 3b, 3c and 4.

OASIS

- **Incidence 0.1–5%.**
- **Risk factors:** nulliparity, **midline episiotomy, persistent OP**, Asian race, ↑ birthweight, **operative vaginal delivery**.
- **Short term:** more blood loss and pain, wound disruption and infection (~7% complications).
- **Long term:** anal incontinence ~2x; **4th > 3rd degree**. Dyspareunia data conflicting.

- **Intrapartum endoanal US** can reveal occult tears but is **not recommended** (ACOG).
- **Next pregnancy:**
 - No anal incontinence after repair → a subsequent vaginal birth does **not** raise the incontinence risk → **vaginal route reasonable**.
 - Recurrence is higher than in multiparas without OASIS, but **similar to primiparas** (low). A **recurrent** OASIS carries substantial incontinence risk.
 - Recurrence factors: **macrosomia, operative vaginal delivery**.
 - **Consider cesarean** with prior postpartum **anal incontinence**, OASIS complications needing **re-repair**, or **psychological trauma**.

Episiotomy

Type	Technique	Notes
Midline	From the fourchette through the perineal body, stopping well short of the EAS; 2–3 cm	↑ OASIS
Mediolateral	From the midline of the fourchette, directed 60 degrees off midline toward the ischial tuberosity	Heals to ~45 degrees after crowning distortion resolves. Pain/dyspareunia similar or ↑ short term. Probably preferred to ↓ OASIS
Lateral	Starts 1–2 cm lateral to the fourchette midline, toward the ischial tuberosity	vs mediolateral: same pain, sexual QoL, OASIS; mediolateral repair faster and uses less suture

- **ACOG: selective, not routine** episiotomy (restrictive use ↓ severe perineal trauma).
- **Indications:** fetal jeopardy (speed), shoulder dystocia, breech, macrosomia, operative vaginal delivery, persistent OP, and imminent significant perineal rupture.
- Rates: ↓ **75%** (1979–2006); **~12%** of US vaginal births in 2012.
- **Timing:** when the head is visible to **4–5 cm** during a contraction. Too early → bleeding; too late → tears not prevented.
- **Analgesia:** existing epidural, bilateral pudendal block, or **1% lidocaine**. EMLA (2.5% lidocaine-prilocaine) needs an hour's application.

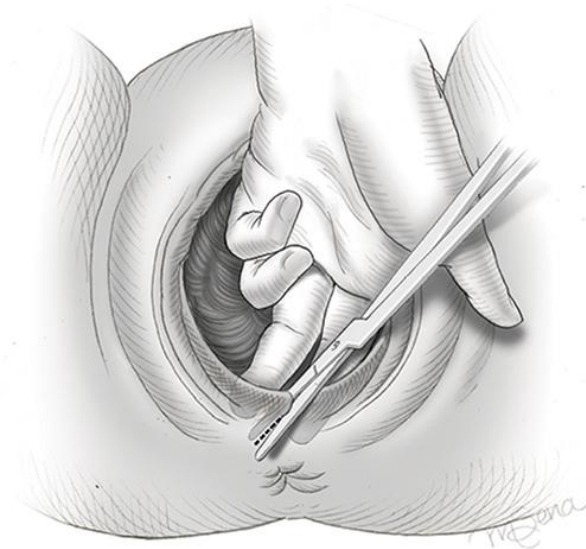


Figure 27-10. Mediolateral episiotomy cut at crowning, 60 degrees off midline toward the ischial tuberosity.

8. Laceration and Episiotomy Repairs

- **Repair after the placenta is delivered** (full attention to separation; no interruption, e.g. for manual removal).
Downside: ongoing bleeding → gauze pressure.

- Adequate analgesia (local lidocaine ± pudendal block; top up epidural). **Reconcile needle and sponge counts** and document.

1st and 2nd degree / episiotomy

- **1st degree:** doesn't always need suturing (only for bleeding or anatomy). Fine absorbable (chromic) or delayed-absorbable (Vicryl, Monocryl); **skin glue** if hemostatic.
- **2nd degree and episiotomy (same steps):**
 1. **Anchor stitch above the apex** → running locking 2-0 closure of vaginal epithelium and deeper tissue, reapproximating the hymeneal ring
 2. Transition stitch to the perineum
 3. **Superficial transverse perineal + bulbospongiosus** reapproximated with continuous nonlocking suture (restores perineal body)
 4. **Running subcuticular** skin closure; final knot proximal to the hymeneal ring
- **Continuous vs interrupted:** continuous is faster with **less short-term pain**; wound complications and long-term pain the same.
- **Suture:** chromic or **2-0 Vicryl**. Vicryl → less short-term pain, but sometimes needs removal → **Vicryl Rapide** reduces this. Also Monocryl, PDS II. Blunt needles ↓ needle-stick injury.

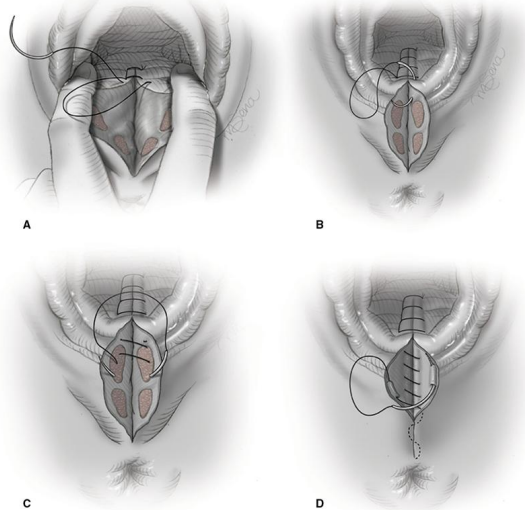


Figure 27-11. Midline episiotomy repair: vaginal closure from above the apex, transition stitch, perineal muscles, then subcuticular skin.

3rd degree (EAS)

- **End-to-end (authors' preference):** retrieve retracted EAS ends; **strength comes from the perisphincter connective tissue ("capsule")** more than the muscle.
 - **4–6 interrupted 2-0 or 3-0 polyglactin 910** sutures at **12, 3, 6 and 9 o'clock**: posterior first, then inferior (6 o'clock), figure-of-eight through the muscle, then anterior and superior.
- **Overlapping:** ends overlapped and sutured with two rows of mattress sutures; **only for 3c** (complete EAS rupture).
- **Neither technique superior** long term.
- **IAS tear:** running nonlocking **3-0 or 4-0**, repaired **before the EAS**.
- **Delayed-absorbable** suture preferred (chromic → ↑ perineal breakdown).

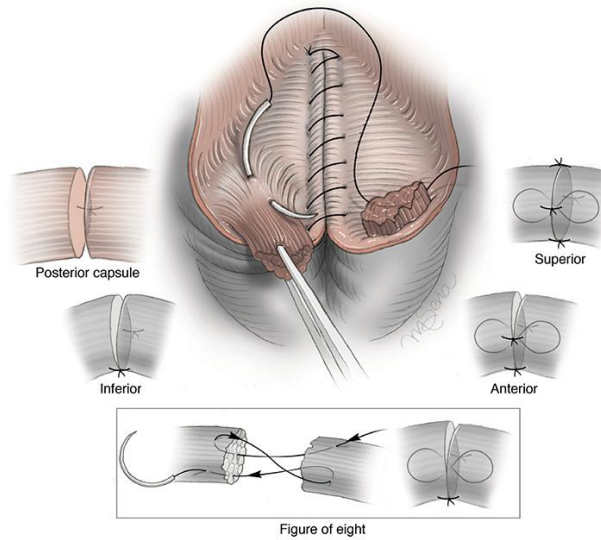


Figure 27-12. End-to-end EAS repair: interrupted sutures through the capsule at posterior, inferior, anterior and superior positions plus a figure-of-eight through the muscle.

4th degree

- **Anorectal mucosa first:** start **1 cm above the apex**, sutures **~0.5 cm apart** in the rectal muscularis (**not entering the lumen**), running **4-0** delayed-absorbable or chromic, down to the anal verge. ± second reinforcing layer.
- Then **IAS** (glistening white layer between submucosa and EAS; often retracted laterally), then EAS.
- **Antibiotics for OASIS:** a **single dose** at repair may be considered: **2nd-generation cephalosporin**, or **clindamycin** if penicillin-allergic.
- **Post-op: stool softeners for 1 week; avoid enemas and suppositories.**

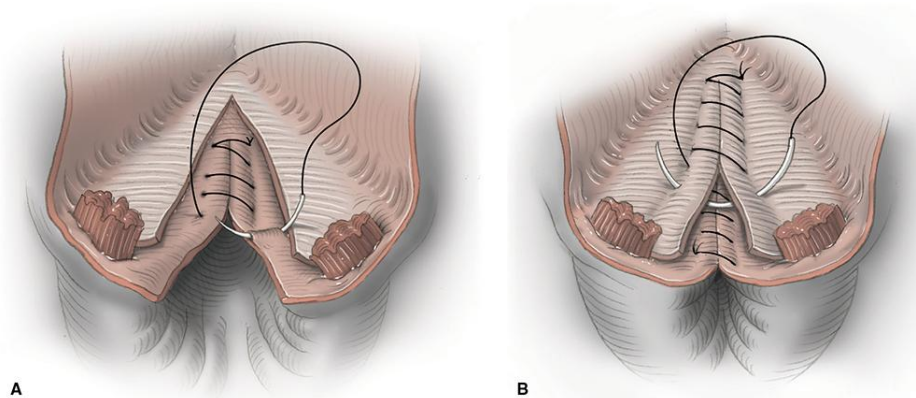


Figure 27-13. Fourth-degree repair: running submucosal anorectal layer (A), then a reinforcing layer incorporating the IAS (B).

Perineal care

- **Ice packs** first (swelling, pain) → later **warm sitz baths**; squirt bottle after voiding/stooling.
- Analgesia: codeine-containing agents; **NSAIDs** for lesser pain.
- ⚠️ **Severe or persistent pain** → examine for **vulvar, paravaginal or ischiorectal hematoma** or **cellulitis** (Ch 37, 42).
- Urinary retention may occur (Ch 36).
- **Intercourse deferred** until after the first postpartum visit (2nd degree/OASIS). Delayed resumption at 3 and 6 months, not at 1 year.

Quick-Recall: Shoulder Dystocia and Third-Stage Essentials

Topic	Remember
Shoulder dystocia definition	Maneuvers needed (ACOG); head-to-body >60 s (Spong)
Incidence / recurrence	~1% / 4–13%
First maneuver	McRoberts + suprapubic pressure (down and lateral)
Doesn't help plexus injury	Episiotomy
Zavanelli tocolytic	Terbutaline 0.25 mg SC
Consider planned cesarean	EFW >5000 g (no DM), >4500 g (DM)
No induction for macrosomia alone	Before 39 wk
Separation signs	Gush, globular fundus, cord lengthening, uterus rises
Active management	Early clamping + controlled cord traction + uterotonic (the key part)
Oxytocin never as	IV bolus (hypotension)
Methylergonovine contraindicated	Hypertension
Misoprostol side effects	Fever, diarrhea
TXA prophylaxis	Not recommended
Midline vs mediolateral	Midline $\uparrow$ OASIS; mediolateral 60° cut $\rightarrow$ heals at 45°
Overlapping EAS repair	Only for 3c
After OASIS repair	Stool softener 1 wk; no enemas/suppositories

Key Numbers to Memorize

- Persistent OP **~5%** (2–10%); nuchal cord **~25%** at term, tight **~6%**
- Lateral neck deviation **$\leq 25\text{--}45^\circ$** during traction
- Delayed cord clamping **30–60 s** (term), **≥ 30 s** (preterm); no cord milking **<29 wk**; clamps **6–8 cm**, cord clamp **2–3 cm**
- Antepartum perineal massage from **34–35 wk**; slow the head once introitus **≥ 5 cm**
- Shoulder dystocia **1%**; brachial plexus injury **11%**, fracture **2%**; HIE **0.5% vs 24%** (<5 vs ≥ 5 min)
- Posterior axilla sling success **96%**; Zavanelli success **91%**; symphysiotomy widens **2.5 cm**
- Home birth neonatal mortality **1.3/1000** (~4 $\times$); water-birth cord avulsion **3/1000**
- Hydrocephaly vaginal delivery if BPD **<10 cm** or HC **<36 cm**
- Placental separation median **4–12 min**; retained placenta **2%**; 90% delivered by **180 / 21 / 14 min** at 20 / 30 / 40 wk; intervene at **15–20 min** (WHO **60 min**)
- Oxytocin **20 U/L at 10–20 mL/min**, then 1–2 mL/min; **10 U IM**; half-life **3–5 min**
- Methylergonovine **200 μ g IM**; misoprostol **400–600 μ g PO**; TXA **1 g IV**
- Vitals: BP/pulse **q15 min $\times$ 2 h**; temp **q4h $\times$ 8 h**
- OASIS **0.1–5%**; anal incontinence **~2 $\times$** ; episiotomy **~12%** of US births; cut at **4–5 cm** head visibility